Delhi Public School, Howrah

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Pre-Board I Examination (2025 – 26)

Class-X

Care must be taken not to write anything on the question paper. All the questions must be attempted at the correct sequence.

Subject: - Mathematics Standard (Code No - 041)

Time: 3 Hours **Maximum Marks: 80**

General Instructions:

Read the following instructions carefully and follow them:

1. This question paper contains 38 questions.

2. This Question Paper is divided into 5 Sections A, B, C, D and E.

3. In Section A, Questions no. 1-18 is multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion-Reason-Based questions of 1 mark each.

4. In Section B, Questions no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.

5. In Section C, Questions no. 26-31 are short answer (SA) type questions, carrying 03 marks each.

6. In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 05 marks each.

7. In Section E, Questions no. 36-38 are case study-based questions carrying 4 marks each with subparts of the values of 1, 1 and 2 marks each respectively.

8. All Questions are compulsory. However, an internal choice in 2 Question of Section B, 2 Questions of Section C and 2 Questions of Section D has been provided. An internal choice has been provided in all the 2 marks questions of Section E.

9. Draw neat and clean figures wherever required.

10. Take π =22/7 wherever required if not stated.

11. Use of calculators is not allowed.

Section A

(Section A consists of 20 questions of 1 mark each.)

1) If x is the LCM of 4, 6, 8 and y is the LCM of 3, 5, 7 and p is the LCM of x and y, then which of the following is true?

   (A) $p = 35x$ (B) $p = 4y$ (C) $p = 8x$ (D) $p = 16y$

2) If the distance between the points $M(\sin\theta, \cos\theta)$ and $N(\cos\theta, -\sin\theta)$ is p, then which of the following statements is true?

   (i) The value of p is 2.
   (ii) The value of p depends on the value of θ.
   (iii) The value of p is independent of θ.
   (iv) The value of p is $\sqrt{2}$.

   (A) (iii) and (iv) (B) (i) and (iii) (C) (ii) and (iv) (D) (ii) and (iii)

3) Consider the equations as shown:

$$(x-a)(y-b) = (x-2a)\left(y-\frac{b}{2}\right)$$

$$x\left(x+\frac{1}{2b}\right) + y\left(y+\frac{a}{2}\right) - 2xy = 5 + (x-y)^2$$

   On comparing the coefficients, a student says these pairs of equations is consistent. Is he/she correct? Which of these explain why?

   (A) Yes, because they represent parallel lines.
   (B) Yes, because they represent intersecting lines.
   (C) No, because they represent parallel lines.
   (D) No, because they represent intersecting lines.

4) A circle has a centre O and radii OQ and OR. Two tangents, PQ and PR, are drawn from an external point P. In addition to the above information, which of these must also be known to conclude that the quadrilateral PQOR is a square?

   (i) OQ and OR are at an angle of 90°. (ii) The tangents meet at an angle of 90°.

   (A) Only (i) (B) only (ii) (C) either (i) or (ii) (D) both (i) and (ii)

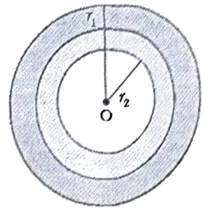
5) If θ is an acute angle and $\tan\theta + \cot\theta = 2$, then the value of $sin^3\theta + cos^3\theta$ is:

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{\sqrt{2}}{2}$ (D) $\sqrt{2}$

6) Consider the equation $kx^2 + 2x = c(2x^2 + b)$. For the equation to be quadratic, which of these **cannot** be the value of k?

(A) 2c (B) 3c (C) 4c (D) 2c + 2b

7) A fountain is enclosed by a circular fence of circumference 11 m and is surrounded by a circular path. The circumference of the outer boundary of the path is 16 m. A gardener increased the width of the pathway by decreasing the area enclosed by the fence such that the length of the fence is decreased by 3 m. The path is to be covered by bricks which cost ₹ 125 per m². What will be the total cost, to the nearest whole number, required to cover the area with the bricks?



(A) ₹ 1,910 (B) ₹ 9,878 (C) ₹ 39,772 (D) ₹ 79,545

8) 2 cards of hearts and 4 cards of spades are missing from a pack of 52 cards. What is the probability of getting a black card from the remaining pack?

(A) $\frac{22}{52}$ (B) $\frac{22}{46}$ (C) $\frac{24}{52}$ (D) $\frac{24}{46}$

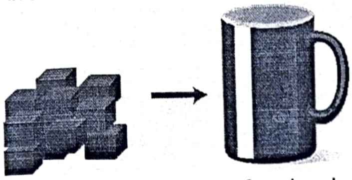
9) If $x\left(\frac{2\tan 30°}{1+tan^2 30°}\right) = y\left(\frac{2\tan 30°}{1-tan^2 30°}\right)$, then $x : y$ =

(A) $1:1$ (B) $1:2$ (C) $2:1$ (D) $4:1$

10) Three bulbs red, green and yellow flash at intervals of 80 seconds, 90 seconds and 110 seconds. All three flash together at 8:00 am. At what time will the three bulbs flash altogether again?

(A) 9:00 am (B) 9:12 am (C) 10:00 am (D) 10:12 am

11) Anishka melted 11 chocolate cubes in cylindrical cup as shown.



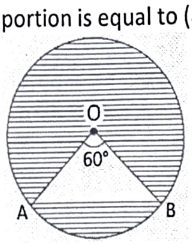
If the length of the side of each cube is k cm and the radius of the cup is r cm, which of these represents the height of the melted chocolate in the cup?

(A) $\frac{7k^3}{4r}$ cm (B) $\frac{7k^3}{2r^2}$ cm (C) $\frac{7k^2}{4r}$ cm (D) $\frac{7k^2}{2r^2}$ cm

12) If one of the zeroes of a quadratic polynomial of the form $x^2 + ax + b$ is the negative of the other, then

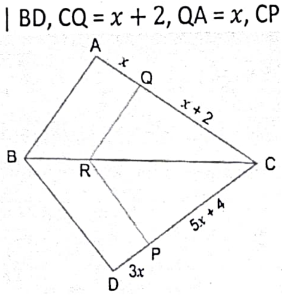
(A) It has no linear term, and the constant term is negative.
(B) It has no linear term, and the constant term is positive.
(C) It can have a linear term, but the constant term is negative.
(D) I can have a linear term, but the constant term is positive.

13) O is the centre of a circle of radius 5 cm. The chord AB subtends an angle 60° at the centre. Area of the shaded portion is equal to (approximately)



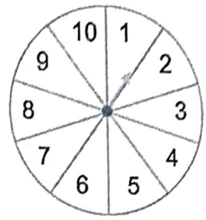
(A) 50 cm^2 (B) 62.78 cm^2 (C) 49.88 cm^2 (D) 67.74 cm^2

14) In the given figure, QR || AB, RP || BD, CQ = $x + 2$, QA = x, CP = $5x + 4$, PD = $3x$. The value of x is



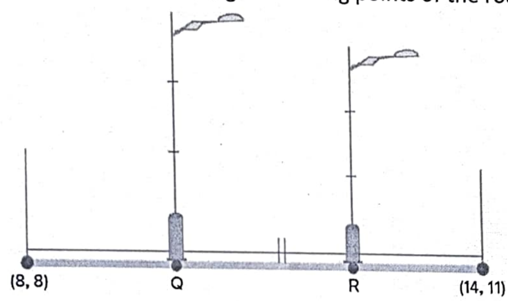
(A) 1 (B) 6 (C) 3 (D) 9

A spinner is shown in the figure.
Some of the events are listed below, when the spinner is spined.
Event-A: The spinner lands on a multiple of 11.
Event-B: The spinner lands on a number less than 11.
Event-C: The spinner lands on a number more than 10.
Which of the following statements is true about the three events?



(A) Probability of event A is 1, so A is a sure event while the probabilities of events B and C are 0, so they are impossible events.
(B) Probability of event B is 1, so B is a sure event while the probabilities of events A and C are 0, so they are impossible events.
(C) Probability of event A is 1, so A is an impossible event while the probabilities of events B and C are 0, so they are sure events.
(D) Probability of event B is 1, so B is an impossible event while the probabilities of events A and C are 0, so they are sure events.

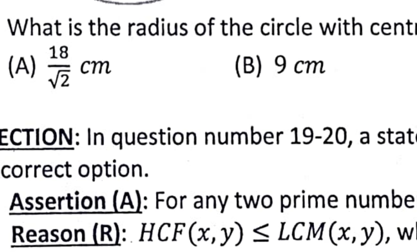
16) Two poles are to be installed on an elevated road as shown in the following figure and both the poles at Q and R trisected the road. The diagram also shows the starting and ending points of the road.



Which of the following are the coordinates of the poles?
(A) $Q(10,9)$ and $R(12,8)$
(B) $Q(10,9)$ and $R(12,10)$
(C) $Q(10,8)$ and $R(12,11)$
(D) $Q(-10,9)$ and $R(0,11)$

17) The mean of n observations is $\bar{x}$. If the first item is increased by 1, second by 2 and so on, then the new mean is:
(A) $\bar{x}+n$
(B) $\bar{x}+\frac{n}{2}$
(C) $\bar{x}+\frac{n+1}{2}$
(D) None of these

18) Two circles with centres O and N touch each other at point P as shown. O, P and N are collinear. The radius of the circle with centre O is twice that of the circle with centre N. OX is a tangent to the circle with centre N, and OX = 18 cm.



(Note: The figure is not to scale.)

What is the radius of the circle with centre N?
(A) $\frac{18}{\sqrt{2}}\,cm$
(B) $9\,cm$
(C) $\frac{9}{\sqrt{2}}\,cm$
(D) $\frac{18}{\sqrt{10}}\,cm$

DIRECTION: In question number 19-20, a statement of **Assertion (A)** is followed by a statement of **Reason (R).** Choose the correct option.

19) **Assertion (A)**: For any two prime numbers x and y $(x < y)$, $HCF(x,y) = x$ and $LCM(x,y) = y$.
Reason (R): $HCF(x,y) \leq LCM(x,y)$, where x, y are any two natural numbers.
(A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, and Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

20) **Assertion (A)**: The value of $\sin\theta = \frac{4}{3}$ is not possible.

Reason (R): Hypotenuse is the longest side in any right-angled triangle.

(A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, and Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Section B

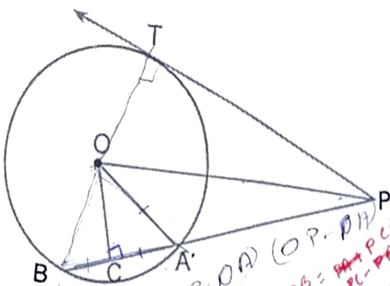
(Section B consists of 5 questions of 2 marks each.)

21) (A) If the last term of an AP of 30 terms is 119 and the 8th term from the end (towards the first term) is 91, then find the common difference of the AP.

OR

21) (B) A sum of ₹ 2,000 is invested at 7% per annum simple interest. Calculate the interests at the end of 1st, 2nd and 3rd year. Do these interests form an AP?

22) In the given figure, PT is a tangent to the circle centred at O. OC is perpendicular to chord AB. Prove that $PA \cdot PB = PC^2 - AC^2$.



(OP+OA)(OP−OA)

PB = PC+AC

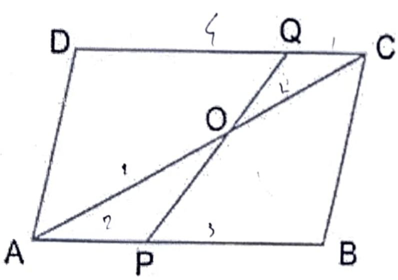
PA = PC−AC

~~PB×PA = (PC+PA)(PC−PA)~~

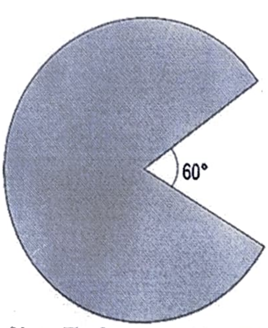
PB×PA = (PC+AC)(PC−AC)

∴ PB.PA = PC² − AC²

23) ABCD is a parallelogram. Point P divide AB in the ratio 2 : 3 and point Q divides DC into the ratio 4 : 1. Prove that OC is half of OA.



24) (A) Wasim made a model of Pac-Man, after playing the famous videogame of the same name. The area of the model is 120π cm². Pac-Man's mouth forms an angle of 60° at the centre of the circle.

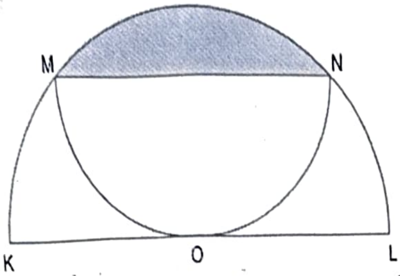


(Note: The figure is not to scale.)

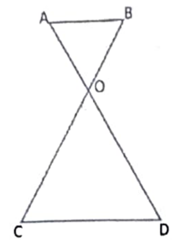
Wasim wants to decorate the model by attaching a coloured ribbon to the entire boundary of the shape. What is the minimum length of the ribbon required in terms of π?

OR

(B) A semicircle MON is inscribed in another semicircle. Radius OL of the larger semicircle is 6 cm. Find the area of the shaded segment in terms of π.



25) The legs of an iron table form two triangles as shown in the picture.

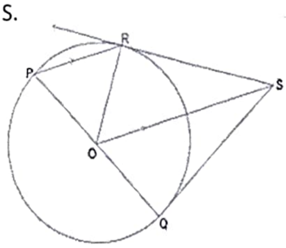



(i) Which similarity criterion is applicable to prove the two triangles similar?
(ii) If AO = 30 cm and OD = 45 cm, then find the ratio of Perimeter of ΔAOB to the Perimeter of ΔCOD.

Section C

(Section C consists of 6 questions of 3 marks each.)

26) In the given figure, PQ is the diameter of the circle with centre O. R is a point on the boundary of the circle, at which a tangent is drawn. A line segment is drawn parallel to PR through O, such that it intersects the tangent at S.



Show that SQ is a tangent to the circle.

27) Prove that $a\sqrt{11} - 3b$ is an irrational number, where a, b are integers and $a \neq 0$.

28) (A) If α, β are the zeroes of the quadratic polynomial $f(x) = x^2 - px + q$. Prove that $\frac{\alpha^2}{\beta^2} + \frac{\beta^2}{\alpha^2} = \frac{p^4}{q^2} - \frac{4p^2}{q} + 2$

OR

28) (B) If the zeroes of the polynomial $x^2 + px + q$ are three times of the zeroes of $2x^2 - 5x - 3$, then find the values of p and q.

29) State and prove Basic Proportionality Theorem.

30 Naima is playing a game using two identical six-sided dice. Each die has three even numbers and three odd numbers on its faces. Before rolling both dice together, she must choose one of the following options to win a prize:

Option I: She wins if the sum of the numbers appearing on the top faces is odd.
Option II: She wins if the product of the numbers appearing on the top faces is odd.
Which option should Naima choose to have a higher probability of winning? Justify your answer with suitable reasoning.

31 30) (A) Given below is a pair of linear equations:

$2x - my = 9$
$4x - ny = 9$

Find at least one pair of the possible values of m and n, if exists, for which the above pair of linear equations has:

(i). A unique solution
(ii). Infinitely many solutions
(iii). No solution

OR

31) (B) A boat goes 30 km upstream and 44 km downstream in 10 hours. In 13 hours, it can go 40 km upstream and 55 km downstream. Determine the speed of the stream and that of the boat in still water.

Section D

(Section D consists of 4 questions of 5 marks each.)

32) In a flight of 2800 km, an aircraft slowed down due to bad weather. Its average speed is reduced by 100 km/h and by doing so, the time of flight is increased by 30 minutes. Find the original duration of the flight.

33) Prove that: $\frac{(1+\cot\theta+\tan\theta)(\sin\theta-\cos\theta)}{(sec^3\theta-cosec^3\theta)} = sin^2\theta cos^2\theta$

34) (A) A tent is in the shape of a cylinder surmounted by a conical top. If the height and radius of the cylindrical part are 3 m and 14 m respectively, and the total height of the tent is 13.5 m, find the area of the canvas required for making the tent, keeping a provision of 26 m^2 of canvas for stitching and wastage. Also, find the cost of the canvas to be purchased at the rate of ₹ 500 per m^2.

OR

34) (B) A right triangle with sides 3 cm and 4 cm is revolved around its hypotenuse. Find the volume of double cone thus generated. (Use $\pi = 3.14$).

35) (A) The following frequency distribution gives the monthly consumption of electricity of 68 consumers of a locality. Find the mean and mode of the data.

Monthly Consumption (in units)	Number of Consumers
65 – 85	4
85 – 105	5
105 – 125	13
125 – 145	20
145 – 165	14
165 – 185	8
185 – 205	4

OR

35) (B) The mean of the following frequency distribution is 62.8 and the sum of all the frequencies is 50. Compute the missing frequency f_1 and f_2.

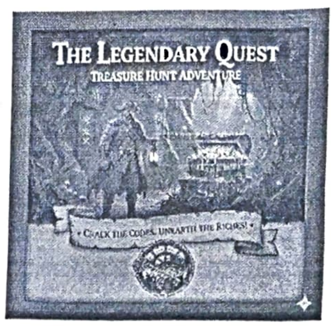
Classes	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100	100 – 120
Frequency	5	f_1	10	f_2	7	8

Section E

(Section E consists of 3 case study-based questions of 4 marks each.)

36) Case Study – 1

Treasure Hunt is an exciting and adventurous game where participants follow a series of clues / numbers / maps to discover hidden treasures. Players engage in a thrilling quest, solving puzzles and riddles to unveil the location of the coveted prize. While playing a treasure hunt game, some clues (numbers) are hidden in various spots collectively forming an AP. If the number on the nth spot is 20 + 4n, then answer the following questions to help the players in spotting the clues.



Based on the above information, answer the following questions:

(a) Which number is on first spot?

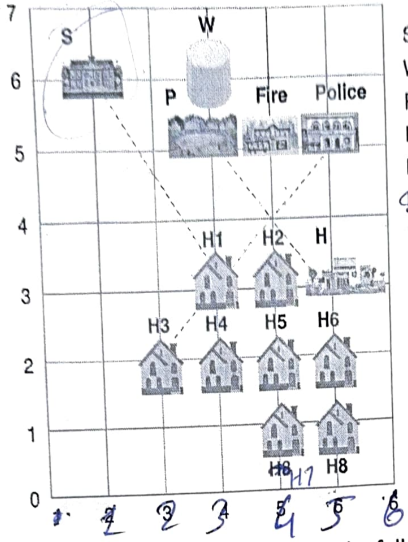
(b) Which number is on the $(n-2)^{th}$ spot?

(c) Which spot is numbered as 112?

OR

(d) What is the sum of all the numbers on the first 10 spots?

37) Shown below is a town plan on a coordinate grid, where 1 unit = 1 km. Consider the coordinates of each building to be the point of intersection of the respective grid lines.



S - School
W - Water Tank
P - Pond
Fire - Fire Station
Police- Police Station
H - Hospital
H1 - House 1
H2 - House 2
H3 - House 3
H4 - House 4
H5 - House 5
H6 - House 6
H7 - House 7
H8 - House 8

Based on the above information, answer the following questions:

(a) Write the coordinates of the hospital.

(b) Write the sum of ordinates of all the houses (House 1 to House 8).

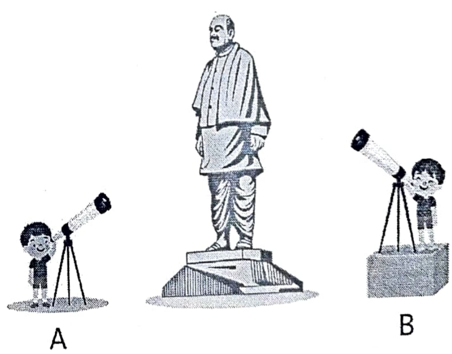
(c) Find the distance between the School and House 1 along the path shown.

OR

(d) What is the ratio in which House 1 divides the path joining House 3 and the Police Station?

38) The statue of Unity situated in Gujarat is the world's largest statue which stands over a 58 m high base. As part of the project, a student constructed an inclinometer and wishes to find the height of the Statue of Unity using it.

He noted the following observations from two places:



Situation – I

The angle of elevation of the top of statue from place A which is $80\sqrt{3}$ m away from the base of the statue is found to be 60°.

Situation – II

The angle of elevation of the top of statue from Place B which is 40 m above ground is found to be 30° and entire height of the statue including the base is found to be 240 m.

Based on the above information, answer the following questions:

(a) Represent the Situation – I with the help of a diagram.

(b) Represent the Situation – II with the help of a diagram.

(c) Calculate the height of the statue excluding the base and find the height including the base with the help of Situation – I.

OR

(d) Find the horizontal distance of Point B (Situation – II) from the statue and the value of $\tan\alpha$, where $\tan\alpha$ is the angle of elevation of top of base of the statue from point B.